COA

25/8/26

Assignment - 1

* Section - A

Ans1. Abstraction is the process of hiding complex implementation details and showing only essential features of a system

- It reduces system complexity
- It makes design and development easier
- Enables modularity

Ans2. Levels of Abstraction:

1. Hardware level - Physical components (CPU, Memory)
2. Machine level - Binary instructions (0s & 1s)
3. Assembly level - Symbolic instructions (ADD, MOV
4. High - level language - C, Java, Python
5. Application level end user - End-user programs.

eg :- Printing "Hello" → Python code → Compiled to assembly → Converted to machine code → executed by hardware.

Ans 3. Functional Units:-

→ Input Unit : Takes input from user
→ Memory Unit : Stores data & instructions
→ Arithmetic logic Unit (ALU): Performs Calculations
→ Control Unit (CU) :- Controls execution ~~of~~ inst.
→ Output Unit :- Displays result

Ans-4: ALU vs CPU:-

- ALU - Performs arithmetic & logical operations
- CPU - Controls and coordinates all operations

Register vs Cache:-

- Registers - Smallest, fastest storage inside CPU
- Cache - Slightly larger, stores frequently used data

Ans-5 Von-Neumann Model:

A complete architecture where CPU, Memory & I/O are connected & instructions/data share the same memory.

Shared-Program Concept:

Instructions are stored in memory like data and executed sequentially.

Ans-6 Limitation caused due to a single path (bus) b/w CPU & Memory.

Key Points:-

- Data and instructions store same bus.
- Causes delay in execution
- Limits system performance

Ans-7 Number of transistors on a chip doubles approximately every 18-24 months.

Impact:-
- Increased Performance
- Reduced Cost
- Smaller & more powerful devices

Ans-8 IPC (Instructions per Cycle) :-

- IPC is the no. of instructions a CPU executes in one cycle
- Higher IPC → Better CPU Performance.

Ans-9 Execution Time :- Total time to run a Program
Clock Rate : No. of cycles per second
CPI (Cycles Per instruction) : Avg cycle per inst
IPC : Instructions Per cycle
Relation :- Execution Time = (Instructions × CPI) / clock Rate

[IPC = 1/CPI]

Ans-10. Definitions

Bit : Smallest unit (0 or 1)
Byte : 8 bits
Signed : Represents positive & negative value
Unsigned : Only +ve values.
eg :- 8-bit unsigned : 0 to 255
8-bit signed : -128 to +127.

Ans11. Basic operations are:-

- Arithmetic (addition, subtraction, multiplication)
- Logical (AND, OR, NOT)
- Comparison (>, <, =)
- Data movement (load | store)

Ans12 Character Representation:

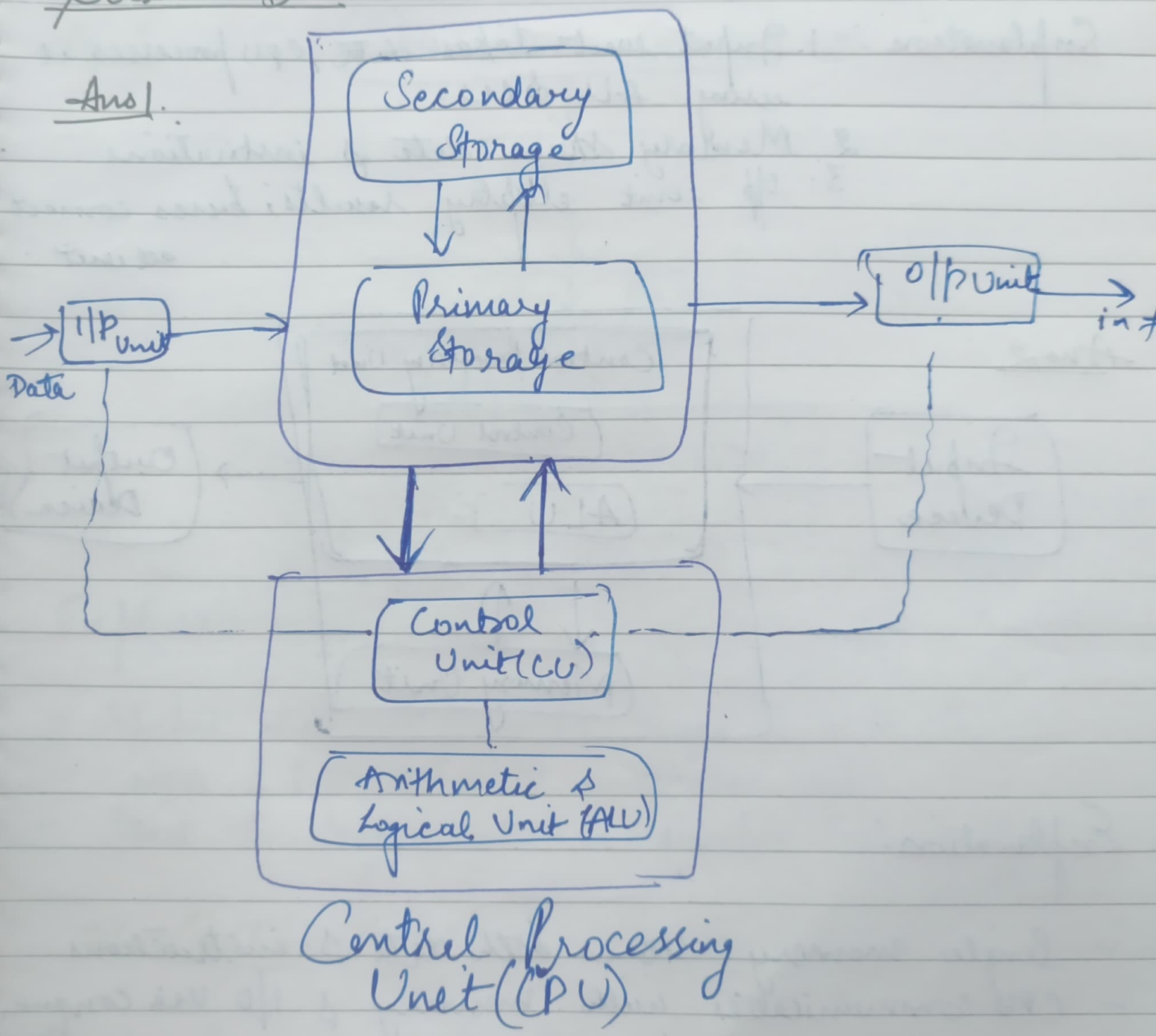
- ASCII (7/8-bit)
- Unicode (supports multiple languages)

Decimal Representation

- fixed-point (simple representation)
- floating point (scientific, eg:- IEEE 754)

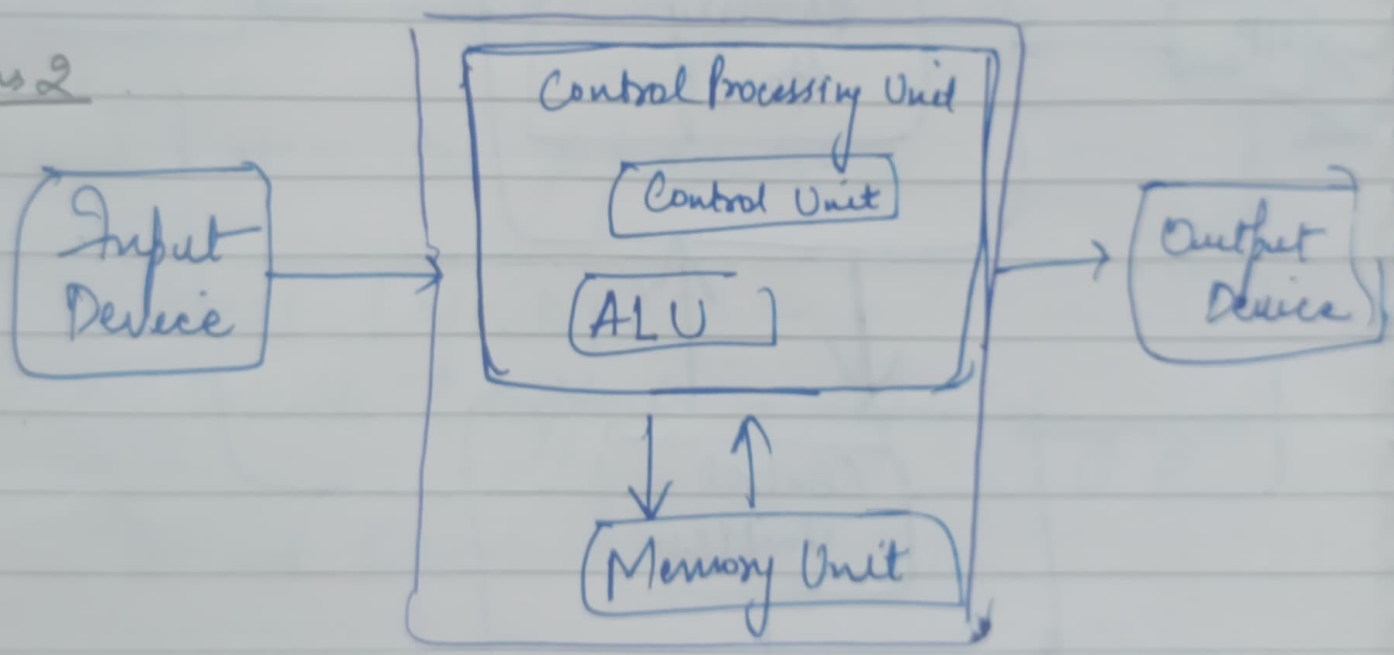
* Section - B

Ans 1.



Explanation :- 1. Input unit takes data, CPU processes it using ALU & CU.
2. Memory stores data & instructions
3. O/P unit display results; buses connect all unit.

Ans 2



Explanation :-

- Single memory stores both data & instructions
- CPU communicates with memory & I/O via common bus.
- Basis of most modern computer.

Ans 3

Instruction Count
↓
CPI
↓
Clock Rate
↓
CPU execution Time

[IPC = 1/CPI

Explanation :-

- 4-bit 2's complement range - 8 to +7
- MSB (leftmost bit) indicates sign
- Used to represent -ve numbers efficiently.

* Section-C

Ans 1 Given, CPU A: 3 GHz, CPI 1.5
CPU B: 2.2 GHz, CPI 1.0

Formula: CPU Time = (Instruction Count X CPI) / Clock Rate

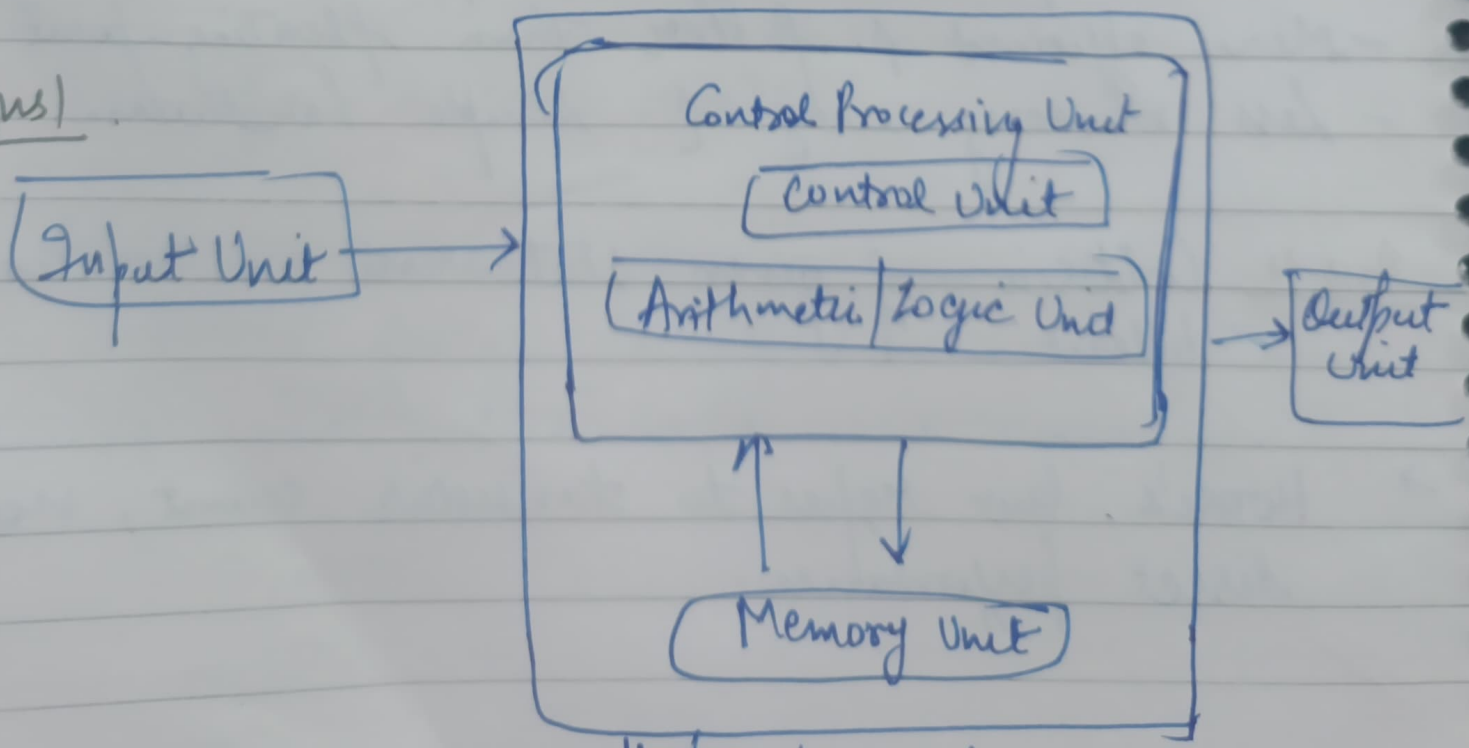
→ Performance gains now limited due to:
- ↳ Power & heat constraints
- ↳ Slower clock speed scaling.

→ Modern improvements depends on:
- ↳ Parallelism (multi-core CPUs)
- ↳ Better architecture

Conclusion, Moore's law alone does not guarantee automatic performance doubling today.

* <u>Section-D</u>

<u>Ans)</u>



Von Neumann Architecture

Design Decision:

- Single memory for data & instructions

Explanation:

- Simplifies system design & reduces hardware cost
- But leads to Von-Neumann bottleneck due its shared bus.

Ans 2 Performance Relation:

CPU Time = (Instruction Count × CPI) / Clock Rate

Explanation:

- Performance improves when CPU time decreases
- Depends on IC, CPI and clock Rate.

One Practical Improvement:

- Reduce CPI using pipeline or better cache
- Reduce IC using optimized algorithms
- Increase clock rate using better hardware.

Ans 3 Represent Number:

Lets take +5 & +4 (4 bit - system)

→ +5 → 0101
→ +4 → 0100

Addition:

$$\begin{array}{r} 0101 \\ +\ 0100 \\ \hline \boxed{1001} \end{array}$$

Interpretation: 1001 in 2's complement = -7.

Overflow Check:

- Adding two positive numbers → result should be +ve
- But result is -ve → overflow occurs.